SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 40**
Question Count: 5
Duration :60 Minutes

Code of Conduct

I **R. Pranathi(192525076)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. Pranathi

Assessment Tasks
Industry Problem Solving Task

1. **Retail Data Optimization Problem -** A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

2. **Social Media Fraud Detection Problem -** A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

3. **Government Data Platform Problem -** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

4. **Banking Fraud Detection Problem** - A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data.
Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

5. **Agriculture Smart Farming Problem** - A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve

yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

1. Retail Data Optimization Problem - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
ANS:

A retail company has both physical stores and an online shopping platform. It produces a large amount of data from sales, inventory, customer purchases, website searches and customer behaviour. The company needs to know the stock available in real time and also wants to predict future product demand. It also wants to give personalized recommendations to customers. For this purpose, a big data architecture can be used.

First, data is collected from different sources such as POS systems, online websites, mobile applications, inventory systems and customer activities. The data may be structured, such as sales records, or unstructured, such as customer reviews.

The collected data is sent to a data ingestion layer. For real-time data, Apache Kafka can be used because it can continuously receive sales and inventory updates. For older or batch data, normal batch data processing can be used.

The data can then be stored in a Data Lake using technologies such as HDFS or cloud storage. A data lake is useful because it can store large amounts of different types of data in one place.

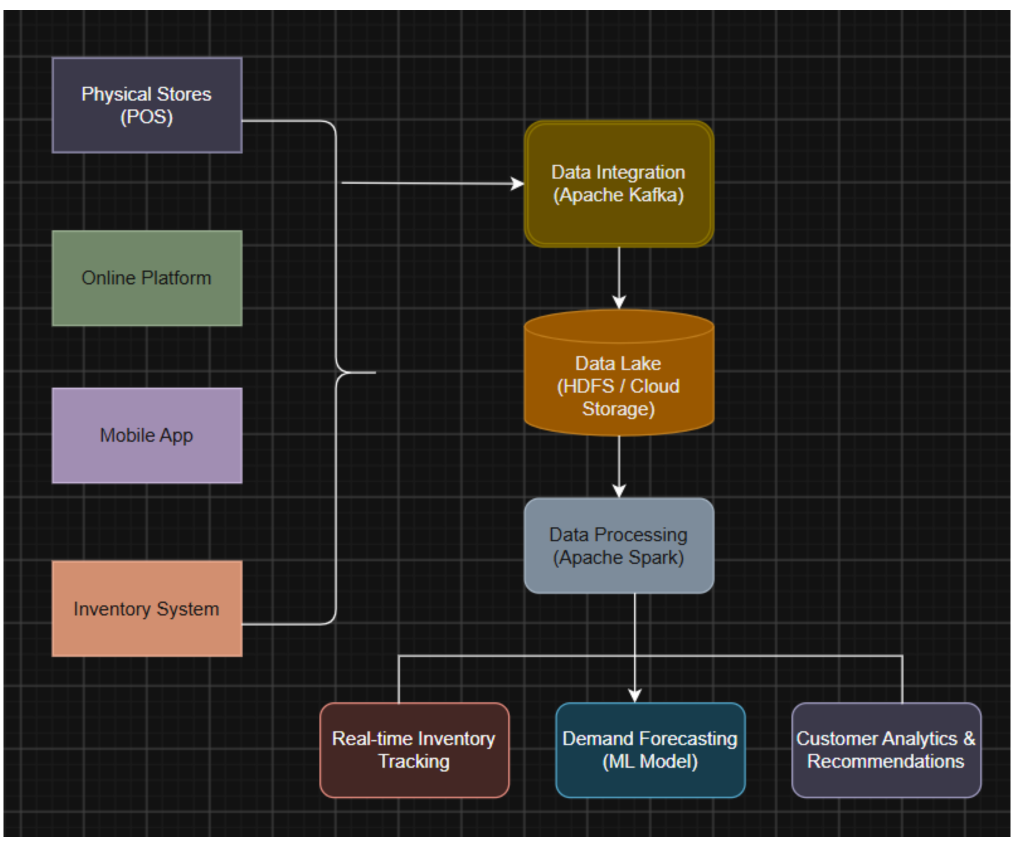
After storing the data, Apache Spark can be used for data processing. Spark can clean the data, remove duplicate records and combine information from different stores and online platforms. Since Spark supports distributed processing, large amounts of retail data can be processed faster.

For real-time inventory tracking, streaming data from POS systems can be processed immediately. For example, if a customer purchases 5 products from a store, the inventory quantity can be updated immediately. The company can also see which products are running low.

For demand forecasting, historical sales data can be analyzed using machine learning models. A forecasting model can study previous sales, seasonal demand and customer behaviour to predict future demand. For example, if the company sells more umbrellas during the rainy season, the system can predict higher umbrella demand and help the company maintain sufficient stock.

For personalized marketing, customer behaviour such as previous purchases, searches, ratings and browsing history can be analyzed. Recommendation models such as collaborative filtering can identify customers with similar interests and recommend suitable products.

The architecture can be represented as:



The main technologies used are Apache Kafka for real-time data ingestion, HDFS or cloud storage for storing big data, Apache Spark for processing, and machine learning models for forecasting and recommendations.

For example, a large retail company can use this system during a festival sale. When thousands of customers purchase products, Kafka receives the transaction information, Spark processes it and the inventory is updated quickly. The forecasting model can also identify products that may soon become out of stock. At the same time, the recommendation system can suggest related products to customers.

This architecture helps the company reduce stock shortages, improve demand planning, provide personalized offers and make better business decisions. It is also scalable because more computing and storage resources can be added when the amount of data increases.

So, A big data architecture combining real-time ingestion, distributed storage, Spark processing, forecasting and recommendation models can provide efficient inventory management and personalized customer experiences for a modern retail company.

2. Social Media Fraud Detection Problem - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
ANS:

A social media platform receives billions of activities every day. These activities include posts, comments, likes, shares, follows and messages. Among these activities, some may come from fake accounts or may contain false information. Therefore, the platform needs a big data system that can process this information quickly and identify suspicious activities.

First, data is collected from different sources such as posts, comments, likes, shares, user profiles and login activities. Since the data is generated continuously, **stream processing** is required. A framework such as **Apache Kafka** can collect and transfer the data in real time.

The incoming data can then be processed using **Apache Spark Streaming** or similar stream-processing technology. It can check user activity continuously instead of waiting until the end of the day. This makes the system suitable for detecting suspicious activities quickly.

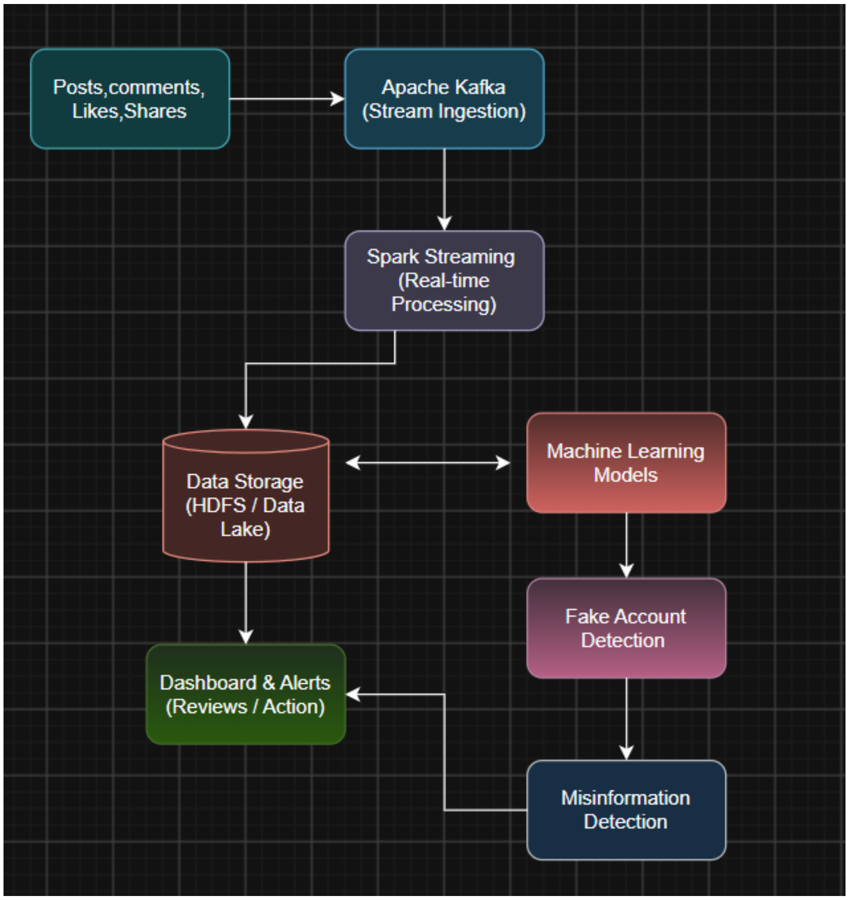
The data can be stored in distributed storage such as **HDFS or cloud storage**. Distributed storage is useful because the platform handles billions of interactions and needs large storage capacity.

Machine learning plays an important role in detecting fake accounts and misinformation. The model can study features such as the number of followers, posting frequency, repeated content, unusual login locations and interaction patterns. Based on these features, it can identify whether an account or activity is suspicious.

For example, if an account is newly created and starts posting hundreds of similar messages and sharing them repeatedly within a few minutes, the machine learning model may give the account a high-risk score.

For misinformation detection, the system can analyze the content of posts and compare their patterns with known unreliable information. Suspicious posts can be sent for further checking or marked for review.

A simple architecture is:



The main advantage of stream processing is **speed**. It allows the platform to identify suspicious behaviour almost immediately. Machine learning improves detection because it can learn patterns from previous fake accounts and misinformation cases.

The system should also be scalable. Since the platform receives billions of interactions, additional processing nodes can be added when the workload increases.

For example, if a large number of fake accounts suddenly start liking and sharing the same post, the streaming system can detect the unusual activity and the machine learning model can identify it as suspicious.

Therefore, combining **Kafka, Spark Streaming, distributed storage and machine learning** provides a scalable solution for detecting fake accounts and misinformation. It improves the speed of detection and helps the social media platform provide a safer environment for users.

3. Government Data Platform Problem - A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
ANS:

A government agency is building a national data platform that combines census data, economic data and geospatial data. Since different government departments need to use the same information, the system should provide common access while maintaining privacy and security. A proper **data governance and data management strategy** is required.

First, all departments should follow common rules for collecting, storing and sharing data. A central **data governance team** can define these rules and decide who can access different types of data. Each department should also have a data owner who is responsible for the quality and security of its data.

The different datasets should be stored in a centralized **data lake or national data platform**. Census, economic and geospatial data can be integrated using common formats and standards. A common data format helps different departments understand and use the information correctly.

Interoperability is important because different government departments may use different software and databases. APIs and common data standards can be used to allow systems to exchange information. Metadata should also be maintained to describe the source, type and meaning of each dataset.

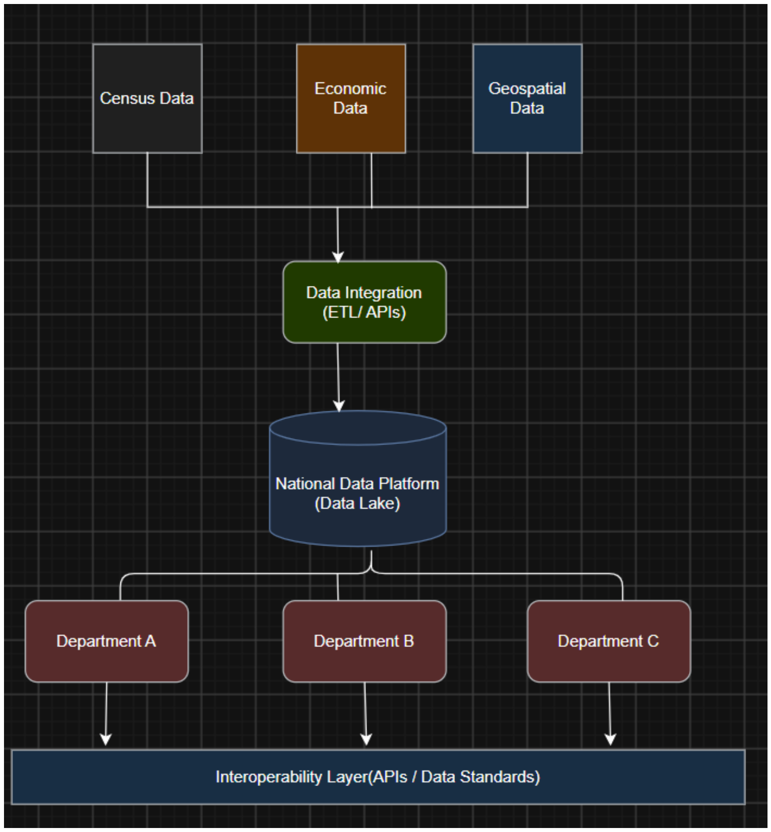
Data quality should be checked before the data is shared. Duplicate, missing and inconsistent data should be identified and corrected. Regular data validation should be performed to maintain accurate information.

Privacy and security are very important because census and economic data may contain sensitive information. **Encryption, authentication, access control and secure APIs** should be used. Only authorized employees should be allowed to access sensitive datasets.

Role-Based Access Control (RBAC) can be used to give different permissions to different departments. For example, one department may only view economic statistics, while an authorized department may access detailed census information.

The system should also maintain **audit logs**. Every important activity such as login, data access, modification and data sharing should be recorded. This helps identify unauthorized access and supports accountability.

A simple structure is:



For example, the planning department can combine population data with economic and geographical data to identify areas that need new schools, hospitals or transport facilities.

Thus, a strong governance strategy with **common standards, data quality management, interoperability, access control, encryption and auditing** can help different government departments safely share and use national data. It improves decision-making while protecting citizen privacy.

4. Banking Fraud Detection Problem: A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data.Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.
ANS:

A bank processes millions of transactions such as UPI payments, ATM withdrawals, credit card payments and online transfers every minute. Among these transactions, some may be fraudulent. Therefore, the bank needs a big data pipeline that can analyze transactions in real time and identify suspicious activities quickly. Since the number of transactions is very high, the system should have **low latency, high accuracy and good scalability**.

First, transaction data is collected from different sources such as mobile banking, ATMs, credit cards, UPI and internet banking. Historical transaction data is also collected because it helps the system understand the normal behaviour of customers.

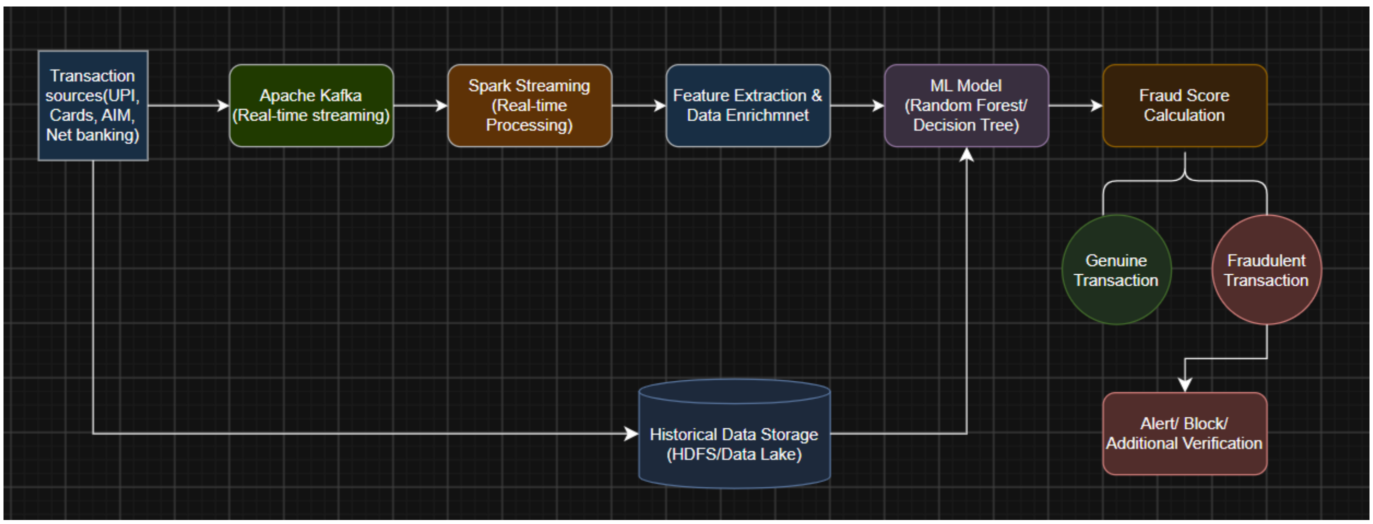
For real-time data, **Apache Kafka** can be used as the streaming platform. Kafka receives transactions continuously and sends them to the processing system. This allows transactions to be analyzed immediately instead of waiting for batch processing.

The incoming data can be processed using **Apache Spark Streaming**. It can combine current transaction information with historical customer information. For example, the system can check the transaction amount, location, device, time and previous transaction pattern.

Machine learning algorithms can then be used for fraud detection. A **Random Forest or Decision Tree** can classify transactions as genuine or suspicious. The model can learn from previous fraudulent transactions and identify similar patterns in new transactions.

For example, if a customer normally makes payments of ₹500–₹5,000 in Chennai but suddenly makes a ₹2 lakh transaction from another country using an unknown device, the system can identify this as unusual activity and give it a high fraud score.

The basic pipeline is:



Historical data can be stored in **HDFS or cloud storage** and used for training the machine learning model. The model can be regularly updated with new fraud cases so that it can recognize new fraud patterns.

Low latency is achieved through stream processing because transactions are analyzed immediately. High accuracy is achieved by combining historical information with machine learning models. Distributed processing also allows the system to handle millions of transactions without depending on a single server.

For example, when a customer makes a large online payment, the system can immediately check the customer's previous transactions, location and device information. If the transaction appears suspicious, the bank can ask for additional authentication or temporarily block it.

Thus, a combination of **Kafka for streaming, Spark Streaming for real-time processing, distributed storage for historical data and machine learning algorithms such as Random Forest or Decision Tree** can provide an effective fraud detection pipeline. It helps the bank detect fraud quickly, reduce financial losses and provide secure banking services.

5. Agriculture Smart Farming Problem - A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.
ANS:

Smart farming uses technology to collect and analyze information about crops and fields. In this system, data is collected from soil sensors, weather sensors, drones and other field devices. Since the data comes from many distributed sources and is generated continuously, a big data solution is useful for managing and analyzing it.

First, soil sensors collect information such as soil moisture, temperature and other soil conditions. Weather sensors provide information about temperature, rainfall, humidity and wind. Drones can capture images of crops and identify crop growth or affected areas.

The collected data is sent to a data ingestion system. For real-time sensor data, Apache Kafka can be used to receive the continuous data. The data can then be stored in distributed storage or cloud storage. This allows large amounts of agricultural data to be stored safely.

Apache Spark can be used to process the collected data. It can combine sensor data, weather information and drone data and identify useful patterns.

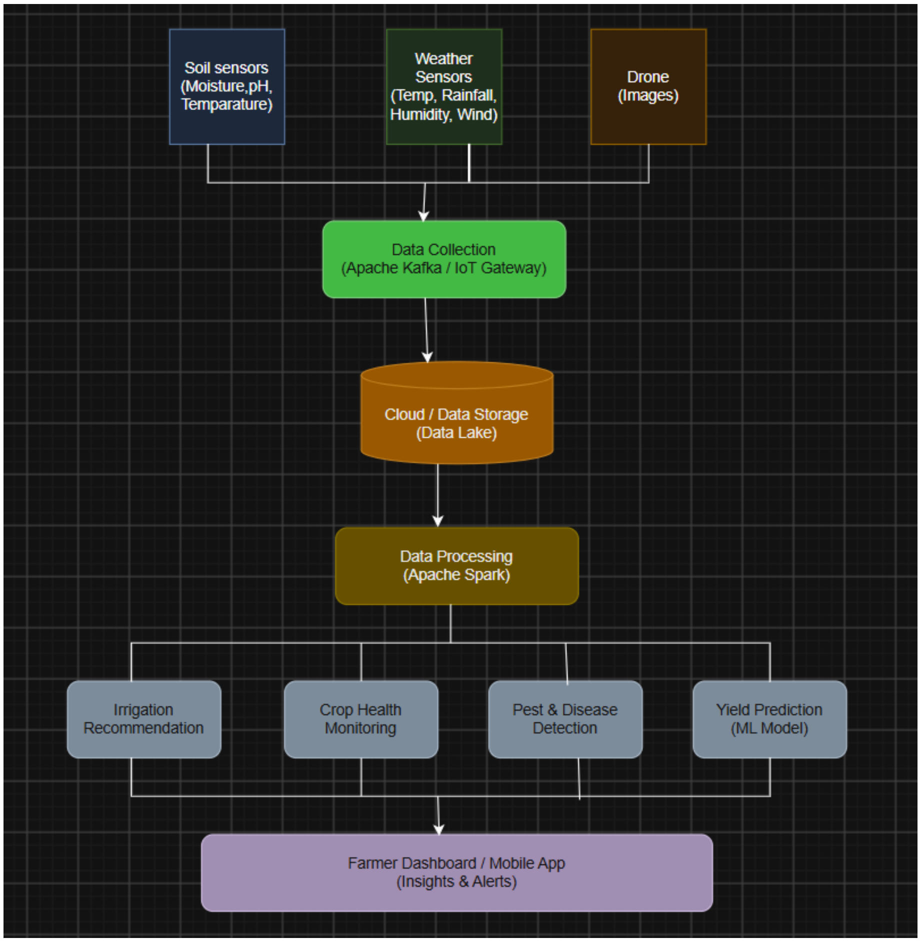
Analytics helps farmers make better decisions. For example, if soil sensors show that a particular field has low moisture, the system can recommend irrigation for that area. This avoids unnecessary water usage.

Weather data can also be analyzed to predict suitable times for irrigation or spraying. If heavy rainfall is expected, farmers can avoid unnecessary irrigation.

Drone images can be analyzed to identify unhealthy crops, pest attacks or areas with poor growth. This allows farmers to take action only in the affected areas instead of treating the entire field.

Machine learning models can also be used for crop yield prediction. Historical crop data, soil conditions and weather information can be analyzed to estimate future crop production.

A simple architecture is:



For example, if sensors detect low soil moisture and the weather system shows no rainfall in the next few days, the system can recommend irrigation. If drone images show that a small part of the field has unhealthy crops, the farmer can treat only that area.

This system helps farmers improve crop yield, reduce water and fertilizer usage, identify crop problems early and make better decisions based on actual field data.

Therefore, sensors, drones, cloud storage, stream processing, Apache Spark and machine learning can be combined to create an effective big data solution for precision agriculture.